

GAS PROCESSING & METHANOL COMPLEX PROJECT NO. 11998A

Project	FEED for Gas Gathering Pipelines & Associated Infrastructure
Client	Brass Fertilizer and Petrochemical Company Limited
Originating Company	Epic Atlantic Limited
Document Title	Civil/Structural MTO
Document Number	EA-BFPCL-GPMC-GGPL-FEED-CIV-LST-001
Document Revision	R01
Document Status	Issued For Review
Originator	Oluseye Akande
Security Classification	Restricted
Issue Date	01-Jul-22

Civil / Structural MTO

R01	01.07.22	Issued for Review	O.Akande	I.Onwuka	W.Irem	
Rev #	Issue Date	Status Description	Originator	Reviewer	Approver (Epic)	Approver (BFPCL)

Revision History

R01	01.07.22	Issued for Review	O.Akande	I.Onwuka			
Rev No	Date	Reason for Issue / Change	Prepared	Reviewed	TCE Approval		

Revision Philosophy:

- a. All documents for review shall be issued at R01, with subsequent R02, R03, etc as required.
- b. All documents approved for issue or approved for design shall be issued at A01 with subsequent A02, A03, etc. as required. (Management of Change is required for A02, A03, etc.).
for the subsequent project phases, post FEED, it is expected that:
- c. All Detailed Design documents for review shall be issued at D01, with subsequent D02, D03, etc. as required.
- d. All documents approved for construction shall be issued at C01 with subsequent C02, C03, etc. as required. (Management of Change is required for C02, C03, etc.)
- e. All approved "As-Built" documents shall be issued at Z01, with subsequent Z02, Z03, etc as required. (Use versions Z01.1, Z01.2, Z01.3, etc to review the "As-Built" document to Z02).

Restricted

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SUMMARY PAGE

Client:	Brass Fertilizer & Petrochemical Company Limited	
Project:	Gas Processing & Methanol Complex	Project No. 11998A
Facility:	Gas Gathering Pipelines & Associated Infrastructure	Doc No. EA-BFPCL-GPMC-GGPL-FEED-CIV-LST-001

The administration of this document is in respect of FEED for Gas Gathering Pipelines & Associated Infrastructure for the Gas Processing & Methanol Plant Project of BFPCL.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2. The focus of the present FEED is on Phase 1, while Phase 2 will be implemented later.

It is also a precursor and accompaniment to the FEED study & report that will lead to the project's FID and implementation.

The report has relied on and used data and reference documents provided by the client, and if further documents and information become available, Epic Atlantic reserves the right to update the opinion set out in this report.

For better appreciation, this document should be read in conjunction with the references listed below, which are the Concept Select Study reports that constitute basis for the Conceptual Study.

KEYWORDS: BASIS OF DESIGN, PIPELINE, MANIFOLD, PLATFORM, GAS, PLANT, ROW, PTS

REFERENCES

Document No.	Description

CIVIL/Structural MTO

DOC. NO. EA-BFPCL-GPMC-GGPL-FEED-CIV-LST-001

Sr. N	Description	Total UoM					Total Weight (kg)	Notes	
		Nr (Pcs)	Length (m)	Area (m2)	Volume (m3)	Unit Weight (kg/m or kg/m2)			
1	W6 X 20		917.8			29.76	27313.7	i. BOUBOUWEI manifold platform	
2	W6 X 25		34			37.2	1264.8		
3	W8 X 40		28			59.2	1657.6		
4	W12 X 45		28			66.9	1873.2		
5	W14 X 22		162			59.5	9639		
6	C8 X 13		24.9			19.3	480.57		
7	1X3/16 SERR GRATTING			1060					
	400 X 200 X 15mm Base								
8	Plate	50							
9	Angle cleat 75x75x10	520							
10	M20 Anchor Bolt	200							
11	CS sh 40 dia 300 pipe		5,000						
12	Concrete grade M-30 to Pile caps and Cased pile				202.6			ii. Indicative for civil MTO iii. Ref Notes on Drawing Nu. EA-BFPCL-GPMC-GGPL-FEED-CIV-DWG-004 iv. Full civil foundation details to be at DED stage	
13	CS Reinforcement bar						16,208		
14	Non-shrink cementious grout				0.3				
15	W8X40		42			59.2	2486.4	i. ELEPA manifold platform	
16	W12X79		1292.8			117.6	152033.28		
17	W14X48		168			71.43	12000.24		
18	W24X94		325			139.8	45435		
19	C8X40		24.9			19.3	480.57		
20	1X3/16 SERR GRATTING			1,673.88					
	400 X 200 X 15mm Base								
21	Plate	52							
22	Angle cleat 75x75x10	336							
23	M20 Anchor Bolt	208							
24	CS sh 40 dia 300 pipe		5,200						
25	Concrete grade M-30 to Pile caps and Cased pile				211.4			ii. Indicative for civil MTO iii. Ref Notes on Drawing Nu. EA-BFPCL-GPMC-GGPL-FEED-CIV-DWG-004 iv. Full civil foundation details to be at DED stage	
26	CS Reinforcement bar						16,912		
27	Non-shrink cementious grout				0.3				